

Participant recruitment criteria (prescreening on Prolific):

1. They have used IoT devices that collect personal information.
 - a. Example of IoT devices:
Smart speaker or voice assistant (e.g., Amazon Echo, Google Home, Apple HomePod),
Smart security system or doorbell camera,
wearable devices (e.g., apple watch, Garmin Fenix, Xiaomi Mi Smart Band, FitBit charge),
Baby monitors,
Smart car,
Smart TV.
(Note: the claim about IoT user can be confirmed with Question1 (in next section) that ask which IoT devices they use. In options we have given only those IoT devices that collect personal data.)
 - b. Example for personal data may include general profile information such as email id, contact, address. Example of specific personal information collected by IoT devices:
 - i. Smart speakers and Smart TVs: voice commands, search queries, music preferences, TV viewing habits
 - ii. Fitness trackers: Heart rate, sleep patterns, calories, location data, exercise routines
 - iii. Connected cars: GPS location, driving habits
 - iv. Smart doorbells: Video footage, visitor patterns, audio recordings
 - v. Baby monitors: audio feed, video feeds, parent response time
2. Participants from EU+EEA+Switzerland. (due to uniform GDPR aligned privacy policies)
3. Fluency in english.
4. Age group: 18 - 100

Introduction to IoT privacy policies study

Thank you for participating in our study on privacy policies in the context of Internet of Things (IoT) devices. Before we begin, please take a moment to read this introduction.

What are IoT devices

IoT, or the "Internet of Things," refers to everyday devices that are connected to the Internet. IoT devices often collect information about you and your environment, to perform specific tasks. These smart devices are designed to make our lives easier, more efficient, or more enjoyable.

Some common examples include: smart watches, fitness trackers, smart speakers, smart home devices for security, baby monitors, Smart TV.

About this study

Our research aims to understand how people interact with privacy policies of IoT devices. Specifically, we want to **explore your impression of privacy policies**:

1. How well you understand privacy policies of IoT devices
2. How easily you can find and use available privacy options in these policies
3. What privacy choices you currently make when using IoT devices
4. What concerns and privacy preferences you have regarding IoT devices

Your responses will help us gain valuable insights into the effectiveness of current privacy policies and user preferences. This information could influence future development of privacy policies, potentially leading to better privacy control and user experiences.

Before you Begin

1. The survey should take approximately 25 minutes to complete.
2. Your responses will be kept confidential and will be analyzed anonymously.
3. There are no right or wrong answers. We're interested in your honest opinions, concerns, and experiences. Your authentic feedback, whether positive or negative, is crucial for the accuracy and value of this research.

Section 1: General Questions

1. Have you ever used, or do you currently use any of the following Internet of Things (IoT) devices? (Select all that apply)
 - a. Smart speaker or voice assistant (e.g., Amazon Echo, Google Home, Apple HomePod)
 - b. Smart home security system or doorbell camera (e.g., Verisure, Google Nest)
 - c. Wearable devices (e.g., fitness tracker such as FitBit, smartwatches such as Apple watch or Samsung Galaxy watch)
 - d. Baby monitors (e.g., LeapFrog, Momcozy, Motorola)
 - e. Smart car, cars that have smart features such as access to real-time traffic information, receive over-the-air updates, adaptive cruise control, (e.g., Tesla, Honda, Toyota)
 - f. Robot vacuum cleaner with sensors (e.g., Roborock, Ecovacs, Philips)
 - g. Augmented Reality / Virtual Reality / eXtended Reality headsets or glasses (e.g., Meta Quest, Apple Vision Pro, Microsoft Hololens)
 - h. Other _____

2. *In a typical week, how frequently do you use any of the IoT devices you listed in previous question?*
 - a. Smart speaker or voice assistant
 - b. Smart home security system
 - c. Wearable devices / fitness trackers
 - d. Baby monitor
 - e. Smart car
 - f. Robot vacuum cleaner
 - g. Augmented Reality/Virtual Reality/eXtended Reality headset or glasses
 - h. Other _____
 - a. Never
 - b. Rarely
 - c. Sometimes
 - d. Often

3. What is the highest level of education you have completed?
 - a. High school diploma or equivalent
 - b. Bachelor ´s degree (e.g., BA, BS)
 - c. Master ´s degree (e.g., MA, MSc, ME)
 - d. PhD degree
 - e. No formal education

Section 2: Awareness of presence of privacy policies

4. Please indicate your level of agreement with the following statements.
- a. I am aware that there is a privacy policy associated with the IoT device which collects personal data.
 - b. I am aware that the primary purpose of a privacy policy is to inform users how a company collects, uses, and shares their personal data.
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- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree
5. Do you usually read the privacy policies associated with your IoT devices.
- a. Yes
 - b. No

Followup question:

Why don't you typically read privacy policies? (Select all that apply)

- a. They are too long to read
 - b. I don't have time to read them
 - c. The language is too complex
 - d. The structure is confusing and hard to follow
 - e. I don't think they're important
 - f. I feel overwhelmed by the number of privacy policies I encounter
 - g. Other reason(s) _____
6. Thinking about your most recent IoT device purchase or setup, how were you primarily informed about its privacy policy? (Select one option that best describes your experience).
- a. The privacy policy was included in the product packaging
 - b. I was prompted to review the privacy policy during device setup or app installation
 - c. I had to actively search for the privacy policy, it wasn't readily available
 - d. I don't recall seeing or being notified about a privacy policy
 - e. Other _____
7. Many IoT devices incorporate **Artificial Intelligence (AI)** capabilities. Examples include, smart speakers using AI-powered voice recognition or smart home devices using AI-powered object recognition.

To what extent do you agree with the following statement:

“Am aware that AI technologies are often mentioned in the privacy policies or product description of IoT devices I own.”

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

Section 3: Comprehension of privacy policies

8. **Personal data** is any information that can be used to identify an individual. Example of Personal data include identifiers such as name, address, email address, identification number, online identifiers such as IP address.

Based on your experience with IoT devices, to what extent do you agree with the following statements regarding their privacy policies:

- a. It is easy to understand **what personal data is collected**.
 - b. It is easy to understand **how personal data is used**.
 - c. I understand the **specific time or duration for which my personal data will be retained or stored**.
- a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly Agree

9. **Personal data** is any information that can be used to identify an individual. Example of Personal data include identifiers such as name, address, email address, identification number, online identifiers such as IP address.

Sensitive data refers to personal information that requires heightened protection due to its nature. Example of sensitive data include biometric data, data concerning health, ethnic origin, philosophical beliefs, political opinion.

NOTE: Sensitive data is a special category of personal data.

Based on your experience with IoT devices, to what extent do you agree with the following statements regarding their privacy policies

- a. I can clearly identify which types of data mentioned are considered **personal data**.
- b. I find it easy to understand the **purpose** for which my personal data are collected and processed.
- c. I can clearly identify which types of data mentioned are considered **sensitive data**.
- d. I find it easy to understand the **purpose** for which my sensitive data are collected and processed.
- e. I can clearly understand **with whom** my personal and sensitive data are **shared** (such as third-parties, business partners).

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

Section 4: Awareness of Privacy Choices

10. **Opt-out** means choosing not to participate or included. Opt-out of certain data collection and sharing means deciding not to allow a company, service, device to collect and share your personal data.

Privacy by default means that when an IoT device or service offers choices about the collection or use of personal data, the most privacy-friendly settings should be the default option.

To what extent do you agree with the following statement:

“I am aware of the *opt-out* option in privacy policies.”

“I am aware of the privacy by default option in privacy policies.”

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

11. When do you typically adjust or select the **privacy by default** option? (Select all that apply).

- a. When I first set up the device
- b. When I hear about security or privacy breaches in the news
- c. When I receive notifications about updates to privacy policy
- d. When I start using a new feature of the device
- e. When someone (friend, family, or expert) recommends it
- f. I have never selected 'privacy by default' option
- g. Other_____

12. **Privacy choice** in a privacy policy refers to **options** that allow you to control how your personal data is collected, used, or shared. Some examples include, **opt-out** of optional cookies, **privacy by default** setting, or **delete** your personal data.

To what extent do you agree with the following statements:

- I. "I am **aware** of the **privacy choices available to me** in the IoT devices I use."
- II. "I would be **interested in receiving personalized recommendations for privacy choices** based on my privacy preferences? "

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

13. **Privacy choice** in a privacy policy refers to **options** that allow you to control how your personal data is collected, used, or shared. Some examples include, **opt-out** of optional cookies, **privacy by default** setting, or **delete** your personal data. (Select one)

How would you prefer these privacy choices to be presented to you?

- a) A dedicated, easy-to-navigate section on available privacy choices
- b) Step-by-step guidance on all privacy choices during initial device setup
- c) Present them as Toggle Controls, with simple on/off switches or sliders to enable or disable specific data collection or sharing practices
- d) I do not have any preference for privacy choice
- e) Other _____

14. In the GDPR, the 'right to access' allows an individuals to find out if an organization is using their personal data. If they do, the individuals can request information about what personal data the organization have about them and how it is used.

To what extent do you agree with the following statements.

“I am aware that I have the **right to access** *for my personal data.*”

“I am aware that I have the *right to access*, and **I have used this right to request information** from organization(s).”

“I wasn't aware of my *right to access* before, but now **I am interested in learning how to use this right.**”

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

Section 5: Privacy behaviors

15. Please indicate how frequently you engage in the following behaviors:

“How often do you *consider the privacy implications before purchasing or using* a new IoT device?”

“How often do you *take steps to protect your privacy*, such as reviewing and adjusting privacy settings?”

“How often do you *accept the pre-configured privacy settings* on your IoT devices? Note that pre-configured privacy settings may not prioritize user privacy and often require manual adjustment.”

- a. Never
- b. Rarely
- c. Sometimes
- d. Often

16. Please indicate to what extent do you agree with the following statements.

“I often **click 'I agree' or 'OK' or 'I Consent'** to accept privacy policies **without reading them thoroughly.**”

“I would be **willing to pay a premium price** for an IoT device that offers **stronger privacy protections**, such as minimal data collection and local data processing.”

“I have **stopped using** an IoT device **due to concerns about its privacy practices.**”

“I have **chosen not to purchase** an IoT device **due to privacy concerns.**”

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

17. Which of the following best describes your attitude towards accepting privacy policies for IoT devices? (Select one)

- a. I accept them without reading because I need to use the device.
- b. I skim through and accept unless something seems off.
- c. I carefully review and only accept policies I fully agree with.
- d. I accept them even if I don't understand, assuming they are designed to protect me and would not cause any harm.
- e. I'm not comfortable accepting policies I don't understand
- f. Other _____

Section 6: Privacy concerns

18. Which type of data collected by IoT devices concerns you the most? (Select all that apply)

- a. Audio recordings (e.g., from smart speakers)
- b. Video footage (e.g., from security cameras, baby monitors)
- c. Location data (e.g., from car, robot vacuum cleaner)
- d. Health and fitness data (e.g., from fitness trackers)
- e. Financial information (e.g., from smart watches)
- f. Other _____
- g. None of these concern me

19. Please indicate to what extent do you agree with the following statements.

“I am comfortable with IoT devices collecting and using my personal data for **targeted advertisement.**”

“I am comfortable with IoT devices collecting and using my personal data for **personalized content.** Example include, personalized news, music, TV shows, movies, or social media feeds.”

“I am comfortable with IoT devices collecting and using my personal data for **sharing it with the device manufacturer.**”

“I am comfortable with IoT devices collecting my personal data and **sharing it with third-parties**, such as advertising partners, data brokers, or other service partners.”

"I am comfortable with IoT devices collecting and using my personal data for **sharing it with government agencies**, such as for law enforcement or other public interest."

"I am comfortable with IoT devices collecting and using my personal data **to improve services and develop new features**."

"Privacy policies of IoT devices **accurately reflect how my personal data is actually used**."

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

20. Please indicate your level of agreement with the following statements:

"I am concerned about my **lack of control over my personal data** collected by IoT devices."

"I am concerned about my **lack of control when** information collected by my IoT devices is **combined with data about me from other sources** (such as, social media, online purchases, public records)."

"I am concerned about my **lack of control over how (artificial intelligence) AI systems process, analyze, and make decisions using my personal data**."

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

21. How do you feel about the *trade-off* between privacy and functionality in IoT devices? (Select one)

- a. I strongly prefer privacy, even if it reduces functionality
- b. I somewhat prefer privacy over functionality
- c. I prefer a balance between privacy and functionality
- d. I somewhat prefer functionality over privacy
- e. I strongly prefer functionality, even if it reduces privacy

Section 7: Privacy preferences

22. In context of this survey, **Privacy by default** means that when an IoT device or service offers choices about the collection or use of personal data, the most privacy-friendly settings should be the default option.

Personal data include information such as name, address, email address, online identifiers such as IP address, location data.

Sensitive data include information such as biometric data, data concerning health, genetic data, political opinions, philosophical beliefs.

NOTE: Sensitive data is a special category of personal data.

Please indicate to what extent do you agree with the following statements.

"I would prefer to have **privacy by default** settings **enabled for personal data** in IoT devices."

"I would prefer to have **privacy by default** settings **enabled for sensitive data** in IoT devices."

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

23. Which of the following privacy options for data control would you prefer to use?
(Select all that apply)

- a. Opt-out of all non-essential cookies
- b. View and download my data
- c. Delete my data
- d. Use right of access
- e. Other _____

24. Which of the following options for privacy settings would you prefer to use?
(Select all that apply)

- a. "**Privacy By default**", get the most privacy-friendly settings enabled automatically
- b. "**Personalized recommendations**", use a digital assistant to receive recommendations for privacy settings based on your privacy preferences
- c. "**Just-in-time**" notices, get notifications when sensitive data is being collected or used
- d. Other(s) _____

25. **Privacy risks** refer to potential negative consequences that may arise from the collection, use, or sharing of your personal data.

These risks may include: unauthorized access to your personal information, Sharing of your data with third parties without your knowledge, Decisions made about you based on inaccurate or incomplete data, etc.

Please indicate to what extent do you agree with the following statements:

“It is important that privacy policies include information about the **privacy risks** associated with processing of my personal data.”

“It is important that privacy policies include information about the **privacy risks** associated with (artificial intelligence) **AI systems** that process my personal data.”

- a. Strongly disagree
- b. Disagree
- c. Neutral
- d. Agree
- e. Strongly Agree

26. Which parts of an IoT device's privacy policy do you care about most?

Please rank the following from 1 (most important) to 5 (least important), by dragging and dropping the items:

- (i) What personal data is collected about me? (e.g., name, contact info, location, browsing history)
- (ii) How is my personal data used? (e.g., to personalize services, for marketing, to improve device functionality)
- (iii) Who is my data shared with? (e.g., third-party companies, advertisers, government agencies)
- (iv) What are my rights regarding my data? (e.g., access, correction, deletion)
- (v) What are my privacy choices? (e.g., privacy settings, opt-out options)

27. For each scenario below, please indicate your level of agreement:

"I would be willing to share my relevant personal data..."

- (i) for a 25% discount on my next purchase
- (ii) for a 70% discount on my next purchase
- (iii) to help create sustainable cities, for example, reducing energy consumption, efficient public transport
- (iv) to contribute to academic research

- (v) during a public health emergency, such as, COVID-19 pandemic
- (vi) during a national security crisis, such as, war

- a. Strongly unwilling
- b. Unwilling
- c. Neutral
- d. Willing
- e. Strongly willing